Aditya Kamath

Exploratory data analysis on Citi bike Data

Introduction:

Citi Bike is New York City's bike share system, and the largest in the nation. Citi Bike launched in May 2013 and has become an essential part of our transportation network. It's fun, efficient and affordable – not to mention healthy and good for the environment.

Citi Bike is available for use 24 hours/day, 7 days/week, 365 days/year, and riders have access to thousands of bikes at hundreds of stations across Manhattan, Brooklyn, Queens and Jersey City.

Citi Bike Trip Histories:

The data includes:

- Trip Duration (seconds)
- Start Time and Date
- Stop Time and Date
- Start Station Name
- End Station Name
- Station ID
- Station Lat/Long
- Bike ID
- User Type (Customer = 24-hour pass or 3-day pass user; Subscriber = Annual Member)
- Gender (Zero=unknown; 1=male; 2=female)
- Year of Birth

Link to Download the Data: https://www.citibikenyc.com/system-data

Description about the dataset:

I have chosen 2 data sets, one for the October 2019 and other for October 2020. Since the data set was very large so I selected data from 1st October in both years and tried to analyse it and get inferences.

There are 15 column variables consisting of different categorical, numerical as well as date time values. The no. of samples in both sets are about 50,000.

I have given colour scheme for the codes:

- Codes with black background are for 2019 data set
- Codes with white background are for 2020 data set.

EDA Using Python:

- **2020**- https://colab.research.google.com/drive/1AcgRAquQUkXLz50eV2vBOs1IEVGfKuDK?usp=sharing
- **2019**- https://colab.research.google.com/drive/1o0argCqMxRfYQVcuf5ugwDyVGRzkf3ji?usp=sharing

EDA Using Tableau:

- https://eda2020.netlify.app/

Problem statement:

- Analyse the public data for present year.
- Make some inference from it.
- Do the same for previous year data.
- See the changes and determine the growth of the chain.
- Also implement certain suggestions for future services.

Objectives:

- ✓ To pre-process the data set.
- ✓ To perform EDA analysis for 2 different data sets.
- ✓ To learn about different kinds of plots in python.
- ✓ To learn about newer techniques (Tableau).
- ✓ To make data visualisation easy.
- ✓ To make it visualisation interactive and easy to access.

Outcomes:

- ✓ Learning Python.
- ✓ Learning Tableau from scratch.
- ✓ Creating a domain and get a hosting service.
- ✓ Hosting the visualisation on a web page for easy access.

Short Note on Tableau:

Tableau Software is an interactive data visualization software company founded in January 2003. It has its free version called Tableau public which offers free access to its software's. It has multiple functionalities and its User Interface is easy enough to visualise the data. Moreover, for larger data set, to perform EDA, we can use tableau as it's an excellent tool.

Aditya Kamath

Steps to do before EDA:

1. Import Packages:

```
import numpy as np
import pandas as pd
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.patches import Circle
import plotly.express as px
%matplotlib inline
```

2. Mount my Drive to colab:

   This step makes it easier to access the file without having to upload the file multiple times.

```
from google.colab import drive
drive.mount('/content/drive')
```

3. Read the file:

```
df1=pd.read_csv('/content/drive/My Drive/EDA/citibike_oct_2020.csv')
df1.head()
```

	tripduration	starttime	stoptime	start station id	start station name	start station latitude	start station longitude	end station id	end station name	end station latitude	end station longitude	bikeid	usertype	birth year	gender
0	222	2020-09-01 00:15:58.6470	2020-09-01 00:19:40.8750	3186	Grove St PATH	40.719586	-74.043117	3276	Marin Light Rail	40.714584	-74.042817	45656	Subscriber	1992	1
1	193	2020-09-01 00:49:00.7370	2020-09-01 00:52:14.3640	3640	Journal Square	40.733670	-74.062500	3206	Hilltop	40.731169	-74.057574	45352	Subscriber	1991	1
2	273	2020-09-01 00:51:44.5870	2020-09-01 00:56:17.7160	3191	Union St	40.718211	-74.083639	3694	Jackson Square	40.711130	-74.078900	44424	Subscriber	1969	1
3	1419	2020-09-01 00:52:48.3160	2020-09-01 01:16:28.0490	3203	Hamilton Park	40.727596	-74.044247	3270	Jersey & 6th St	40.725289	-74.045572	42240	Subscriber	1986	1
4	254	2020-09-01 00:56:16.7170	2020-09-01 01:00:31.0870	3194	McGinley Square	40.725340	-74.067622	3679	Bergen Ave	40.722104	-74.071455	42434	Subscriber	1992	1

4. Get the shape of data frame:

   This is important as we want to know how many values are there in total.

```
df1.shape
(49244, 15)
```

2019

```
df1.shape
(53833, 15)
```

2020

5. We need to check if there are any null values:

   By using .info(), we get to know the columns data type and the number of non-null values in each column. If there is any null value, we need to either drop it or assign them mean values.

```
Data columns (total 15 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   tripduration             49244 non-null  int64
 1   starttime                49244 non-null  object
 2   stoptime                 49244 non-null  object
 3   start station id         49244 non-null  int64
 4   start station name       49244 non-null  object
 5   start station latitude   49244 non-null  float64
 6   start station longitude  49244 non-null  float64
 7   end station id           49244 non-null  int64
 8   end station name         49244 non-null  object
 9   end station latitude     49244 non-null  float64
 10  end station longitude    49244 non-null  float64
 11  bikeid                   49244 non-null  int64
 12  usertype                 49244 non-null  object
 13  birth year               49244 non-null  int64
 14  gender                   49244 non-null  int64
```

```
Data columns (total 15 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   tripduration             53833 non-null  int64
 1   starttime                53833 non-null  object
 2   stoptime                 53833 non-null  object
 3   start station id         53833 non-null  int64
 4   start station name       53833 non-null  object
 5   start station latitude   53833 non-null  float64
 6   start station longitude  53833 non-null  float64
 7   end station id           53833 non-null  int64
 8   end station name         53833 non-null  object
 9   end station latitude     53833 non-null  float64
 10  end station longitude    53833 non-null  float64
 11  bikeid                   53833 non-null  int64
 12  usertype                 53833 non-null  object
 13  birth year               53833 non-null  int64
 14  gender                   53833 non-null  int64
```

6. Convert time:

As we can see that we have a time column which needs to be converted from object to date time. We use NumPy to convert the data.

```
# covert time to int64 (prev it was object)
df1['starttime'] = pd.to_datetime(df1['starttime'])
df1['stoptime'] = pd.to_datetime(df1['stoptime'])
```


```
# covert time to int64 (prev it was object)
df1['starttime'] = pd.to_datetime(df1['starttime'])
df1['stoptime'] = pd.to_datetime(df1['stoptime'])
```


After doing all of this, our data is ready for EDA analysis.

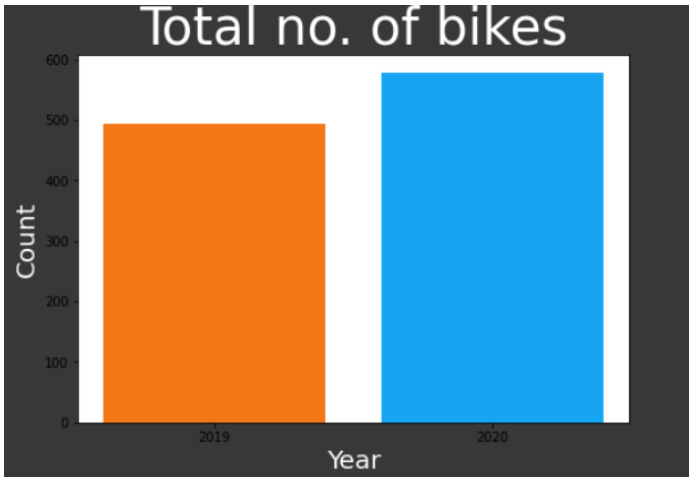
Exploratory Data Analysis

1. How many bicycles are there in the system?

For:

- ✓ 2019: 494
- ✓ 2020: 579

We can Infer that there is Increase no. of cycles from 2019 to 2020.



2. How many stations are there in total?

Start Stations:

- Start Stations: In both cases the total no. of station is 51.
- All station names are given a particular station id and we can confirm this from finding no. of unique station ID.

```
['Journal Square' 'McGinley Square' 'Essex Light Rail' 'City Hall'
 'Paulus Hook' 'Newport Pkwy' 'Marin Light Rail'
 'Columbus Dr at Exchange Pl' 'Harborside' 'York St' 'Newark Ave'
 'Grove St PATH' 'Newport PATH' 'Hamilton Park' 'Manila & 1st'
 'Columbus Drive' 'JC Medical Center' 'Lafayette Park' '5 Corners Library'
 'Glenwood Ave' 'Union St' 'Lincoln Park' 'Oakland Ave'
 'Liberty Light Rail' 'Montgomery St' 'Grand St' 'Baldwin at Montgomery'
 'Warren St' 'Jersey & 6th St' 'Morris Canal' 'Astor Place'
 'Communipaw & Berry Lane' 'Sip Ave' 'Pershing Field' 'Brunswick St'
 'Dey St' 'Monmouth and 6th' 'Bergen Ave' 'Washington St' 'Van Vorst Park'
 'Brunswick & 6th' 'Fairmount Ave' 'Dixon Mills' 'Jersey & 3rd' 'Hilltop'
 'Heights Elevator' 'Hoboken Ave at Monmouth St' 'Riverview Park'
 'Leonard Gordon Park' 'Jackson Square' 'Christ Hospital']
```

2019

```
['Grove St PATH' 'Journal Square' 'Union St' 'Hamilton Park'
 'McGinley Square' 'Baldwin at Montgomery' 'Christ Hospital'
 'Newport Pkwy' 'Jersey & 3rd' 'Astor Place' 'Dixon Mills'
 'Essex Light Rail' 'Marin Light Rail' 'Warren St' 'Fairmount Ave'
 'Lincoln Park' 'Lafayette Park' 'Columbus Drive' 'Monmouth and 6th'
 'JC Medical Center' 'Washington St' 'Jersey & 6th St' 'Morris Canal'
 'Newport PATH' 'Liberty Light Rail' 'Sip Ave' 'Van Vorst Park'
 'Glenwood Ave' 'Brunswick St' 'Brunswick & 6th' 'City Hall'
 'Pershing Field' 'Columbus Dr at Exchange Pl' 'Communipaw & Berry Lane'
 'Hilltop' 'Harborside' 'Paulus Hook' 'Bergen Ave' 'Grand St'
 'Heights Elevator' 'York St' 'Newark Ave' 'Montgomery St'
 'Hoboken Ave at Monmouth St' 'Manila & 1st' 'Dey St' 'Oakland Ave'
 'Jackson Square' '5 Corners Library' 'Riverview Park'
 'Leonard Gordon Park']
```

2020

End Stations:

- End Stations: In both cases the total no. of station is different.
- We can see that in:
  - ✓ 2019- no. of stations 68.
  - ✓ 2020- no. of stations is 72.
- We can infer from this that 4 new stations have been added

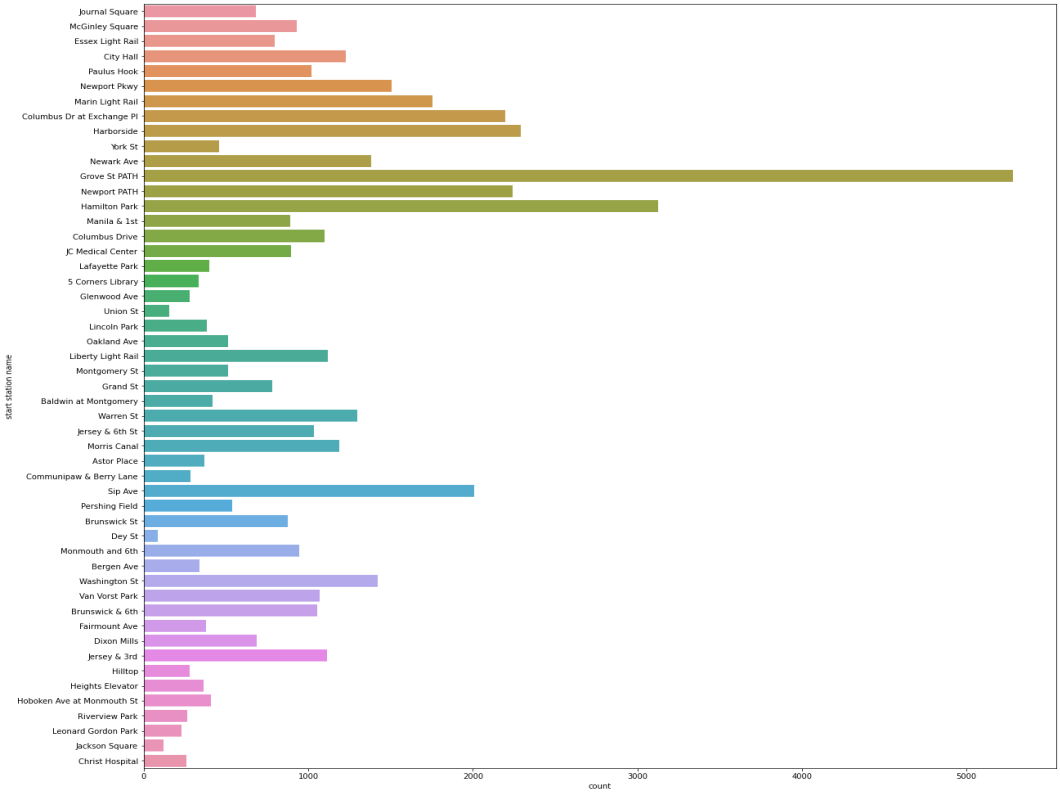
```
['Hilltop', 'Bergen Ave', 'Paulus Hook',
 'Columbus Dr at Exchange Pl', 'Astor Place', 'Essex Light Rail',
 'Warren St', 'Newport Pkwy', 'Union St', 'Newark Ave',
 'Harborside', 'Hamilton Park', 'Liberty Light Rail',
 'Riverview Park', 'Grove St PATH', 'Hoboken Ave at Monmouth St',
 'Manila & 1st', 'Marin Light Rail', 'JC Medical Center',
 'Newport PATH', 'York St', 'Leonard Gordon Park', 'Jackson Square',
 'Oakland Ave', 'Van Vorst Park', 'Jersey & 6th St',
 'Baldwin at Montgomery', 'Journal Square', 'Glenwood Ave',
 'Sip Ave', 'City Hall', 'Brunswick St', 'Communipaw & Berry Lane',
 'Fairmount Ave', 'Columbus Drive', 'Grand St', 'Washington St',
 'Jersey & 3rd', 'Montgomery St', 'Dey St', 'Morris Canal',
 'Pershing Field', 'Dixon Mills', 'Brunswick & 6th',
 '5 Corners Library', 'Monmouth and 6th', 'McGinley Square',
 'Lafayette Park', 'Lincoln Park', 'Heights Elevator',
 'Christ Hospital', 'JCBS Depot', 'Greenwich St & W Houston St',
 '12 Ave & W 40 St', 'West St & Chambers St', 'W 17 St & 8 Ave',
 'Barrow St & Hudson St', 'W 20 St & 11 Ave', 'Jay St & York St',
 'W Broadway & Spring Street', 'Broadway & W 56 St',
 'West Thames St', 'Vesey Pl & River Terrace', 'E 16 St & 5 Ave',
 'Christopher St & Greenwich St', 'Murray St & West St',
 'Mercer St & Spring St', 'Fulton St & Broadway'], dtype=object)
```

2019

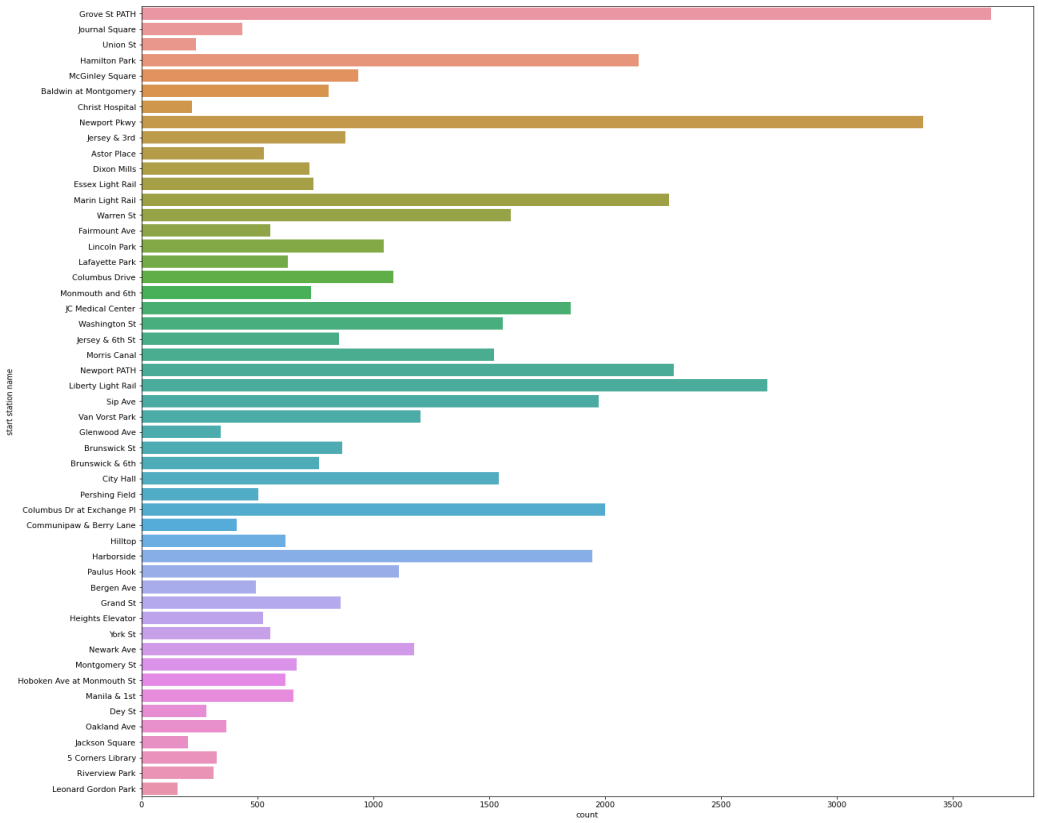
```
['Marin Light Rail', 'Hilltop', 'Jackson Square', 'Jersey & 6th St',
 'Bergen Ave', 'Newport Pkwy', 'Journal Square', 'Christ Hospital',
 'Baldwin at Montgomery', 'Grove St PATH', 'Sip Ave',
 'Columbus Drive', 'Paulus Hook', 'JC Medical Center',
 'Hoboken Ave at Monmouth St', 'Dey St', 'Essex Light Rail',
 'Manila & 1st', 'City Hall', 'Fairmount Ave',
 'Columbus Dr at Exchange Pl', 'Van Vorst Park', 'Lincoln Park',
 'Newport PATH', 'McGinley Square', 'Glenwood Ave',
 'Pershing Field', 'Morris Canal', 'Jersey & 3rd',
 'Liberty Light Rail', 'Washington St', 'Grand St', 'Hamilton Park',
 'Brunswick St', 'Dixon Mills', 'Newark Ave', 'Harborside',
 'Warren St', 'Brunswick & 6th', 'Astor Place', 'Montgomery St',
 'Communipaw & Berry Lane', 'Heights Elevator', 'Oakland Ave',
 'Monmouth and 6th', '5 Corners Library', 'York St', 'Union St',
 'Riverview Park', 'Lafayette Park', 'JCBS Depot',
 'Leonard Gordon Park', 'Broadway & W 49 St', 'West Thames St',
 'W 45 St & 8 Ave', 'Cabrini Blvd & W 177 St',
 'South St & Gouverneur Ln', 'W 13 St & 7 Ave',
 'Mott St & Prince St', 'Front St & Jay St',
 'Riverside Dr & W 78 St', 'Mercer St & Spring St',
 '12 Ave & W 40 St', 'Little West St & 1 Pl',
 'Riverside Dr & W 138 St', '6 Ave & W 33 St',
 'Harrison St & Hudson St', 'Christopher St & Greenwich St',
 'E 85 St & York Ave', 'West St & Liberty St', 'Wall St & South St',
```

2020

3. All Start Stations:

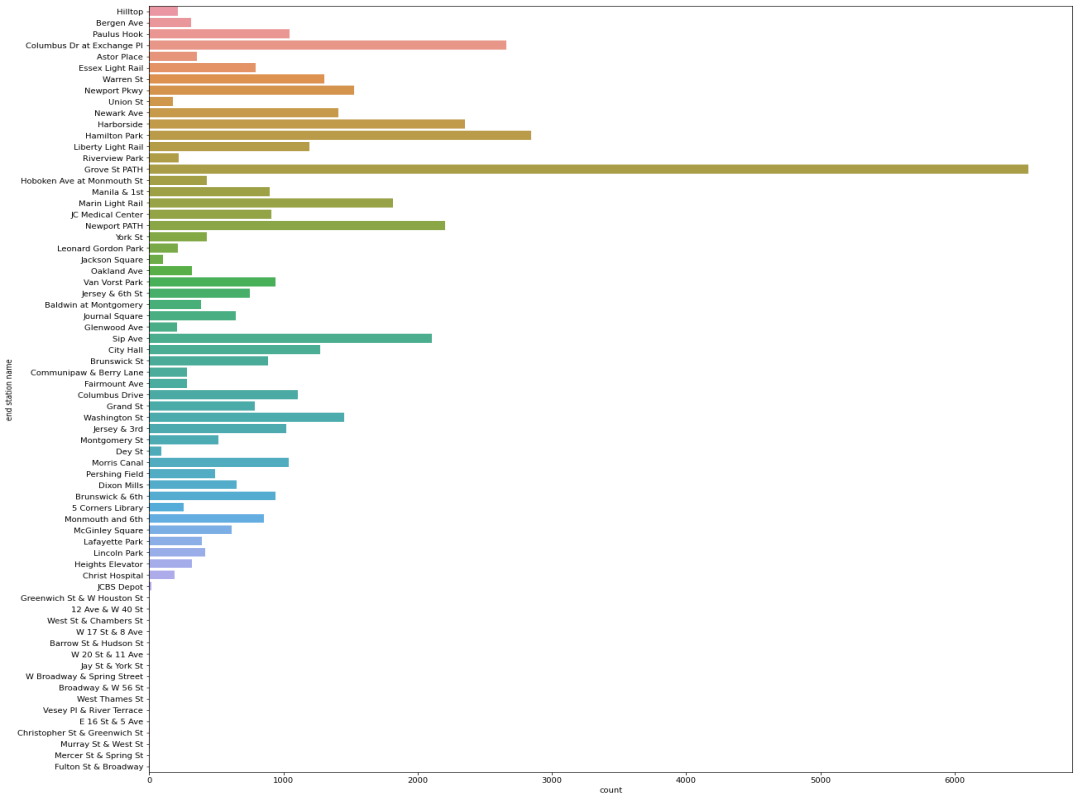


2019

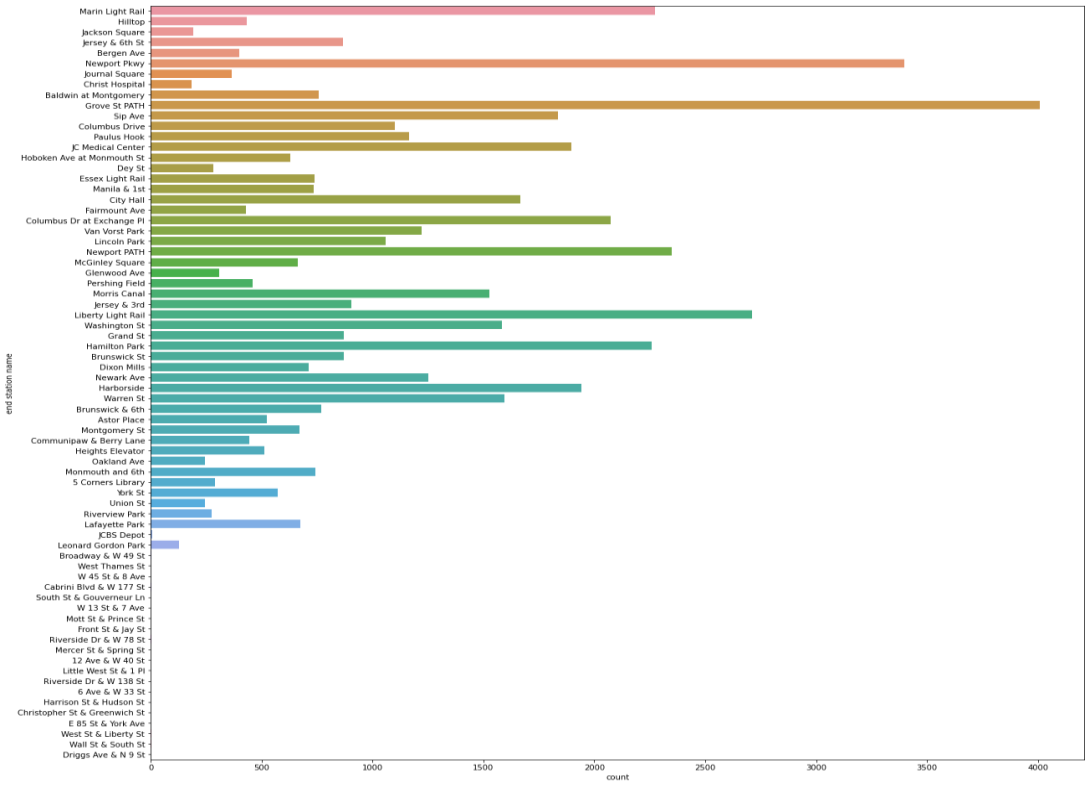


2020

4. All End Stations:



2019



2020

5. Why not all end stations?

	start station id	end station id
3186	5282.0	6548
3203	3124.0	2842
3639	2292.0	2354
3202	2241.0	2206
3792	2196.0	2658
...	...	...
358	NaN	2
363	NaN	1
426	NaN	1
459	NaN	1
468	NaN	1

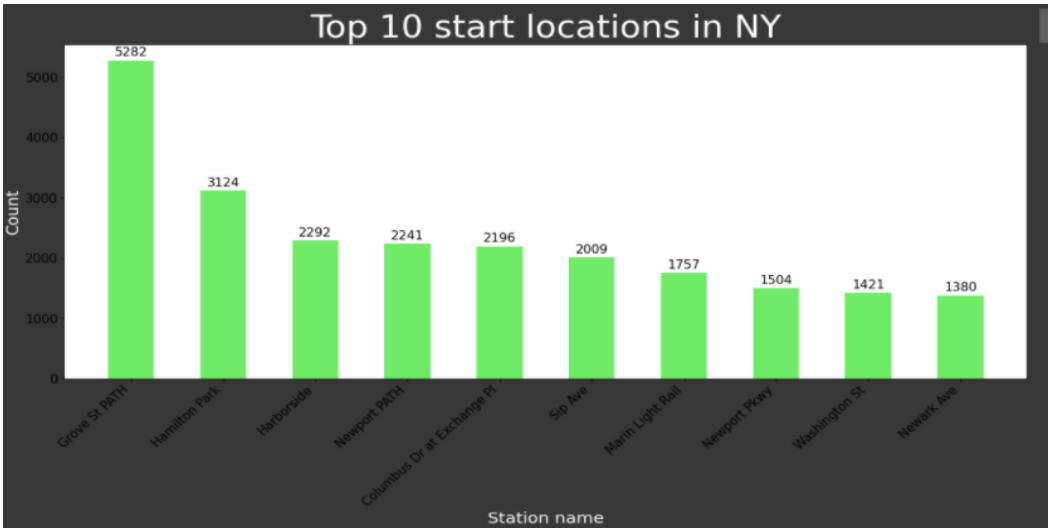
63 rows × 2 columns

	start station id	end station id
3186	3664.0	4009
3199	3371.0	3396
3192	2701.0	2710
3202	2295.0	2349
3276	2276.0	2275
...	...	...
3804	NaN	1
3906	NaN	2
3919	NaN	2
4007	NaN	1
4069	NaN	2

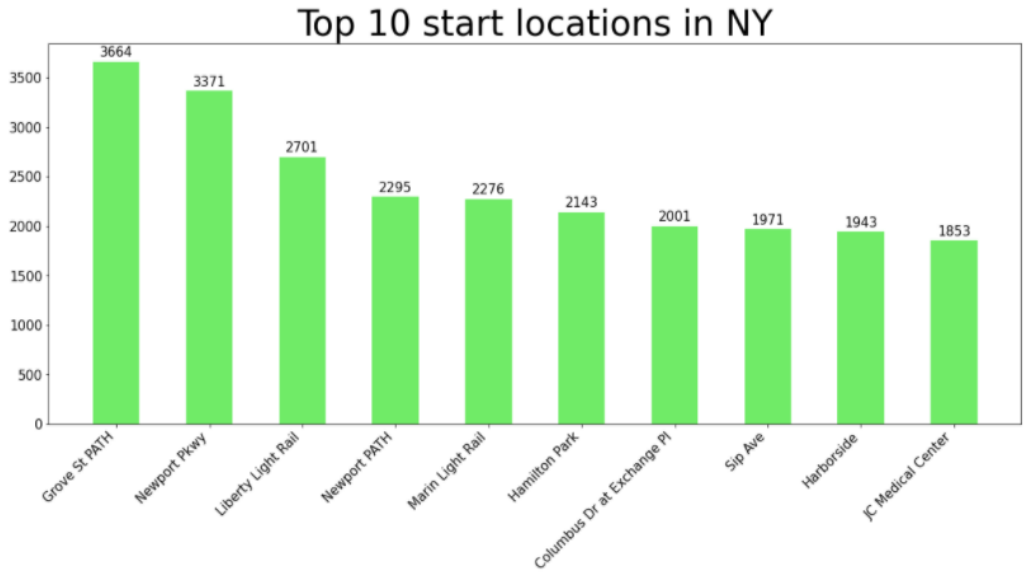
72 rows × 2 columns

- Here we can see the station id and no. of people from a particular station starting and ending
- We can also see the busiest station is one with id 3186
- In general, for each station, the frequency of start station is simliar to that of end station.
- People might make a round trip with citi bike.
- However, there are also stations where people never start a trip but made an end to that point.

6. Top 10 Start Stations:
    - We can infer that the busy start stations have changed from the past year.
    - Grove St. Path still continues to be the busiest station.
    - It is a popular tourist attraction destination. There are many buildings where famous shows like FRIENDS were shot.
    - Hamilton park was a popular sight for spending evenings in 2019 but in 2020 we can see that its nowhere in the top 10 start station list. It might be because users stopped going to the park due to covid 19. Also, news state that the reopening of elementary schools in September again gave rise to no. of cases so the area was again affected.



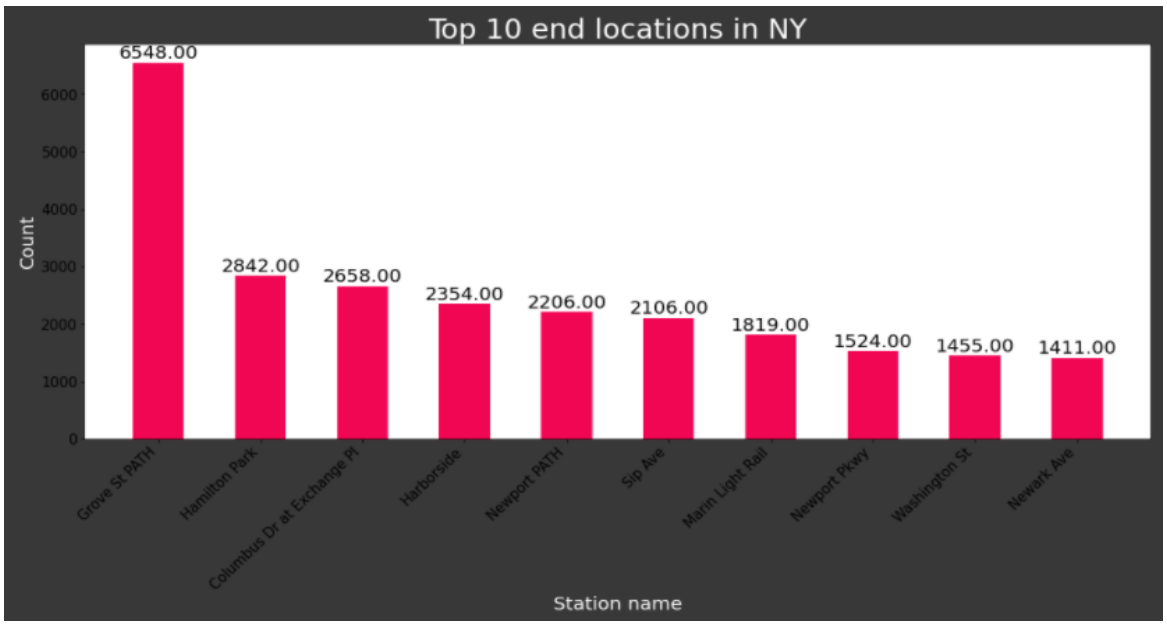
2019



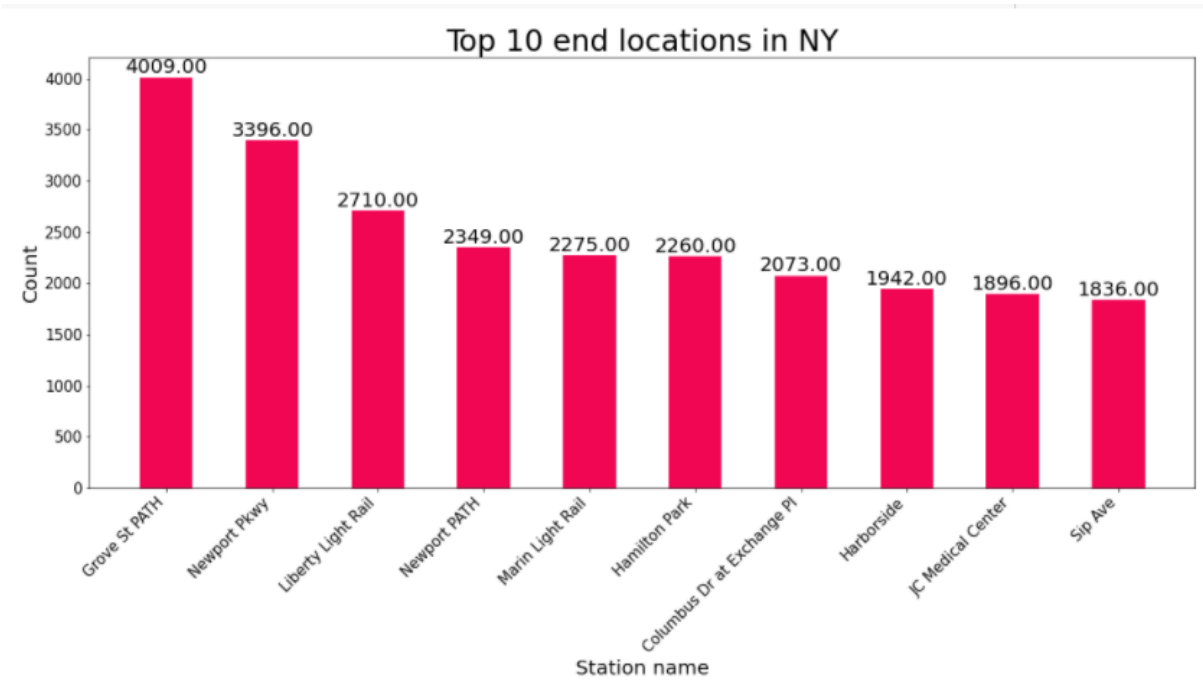
2020

- The ones we see above are the stations which are located mostly on the shore side and/or have popular tourist attraction places where all people can spend some leisure time.

7. Top 10 End Stations:



2019

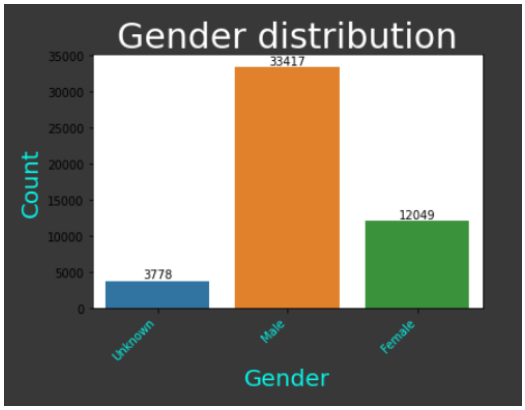


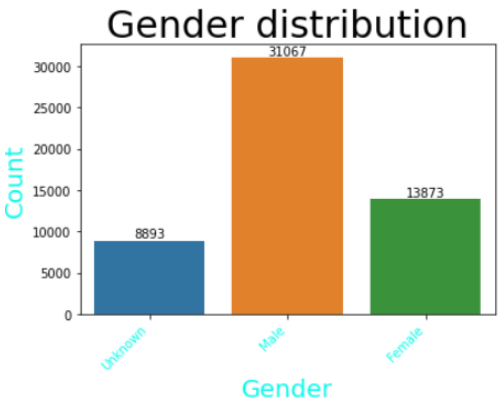
2020

With both the data we can confirm that grove street is the busiest of all stations.

8. Gender Distribution:

- We can infer that the total no. of users is more in previous year and have declined this year.
- Males use the service more than females and unknowns.

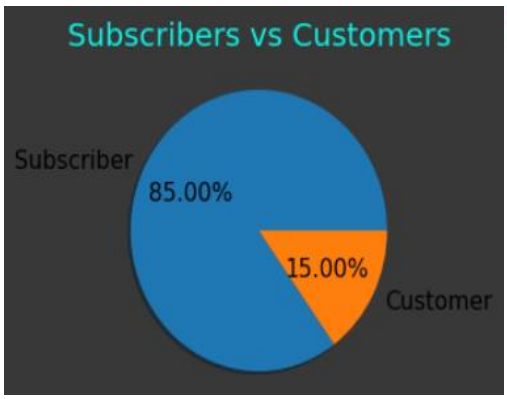




2019 2020

9. Subscriber VS Customer count:

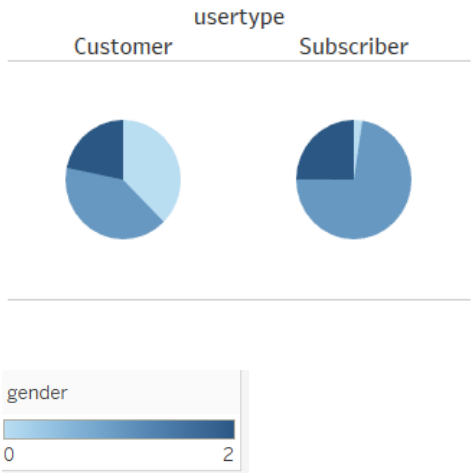
- We can infer that, the customers (mostly tourists) were about 15% in 2019 but in 2020 the number of customers has increased and less people are subscribing.
- It might be because more and more people are concerned about the lockdown and don't want to subscribe for a month and would only renew when and where related.
- This might also be related to the job losses.
- Although it's a pandemic, people still using the service means that they want to remain fit or use cycle to commute to work instead of using public transports.



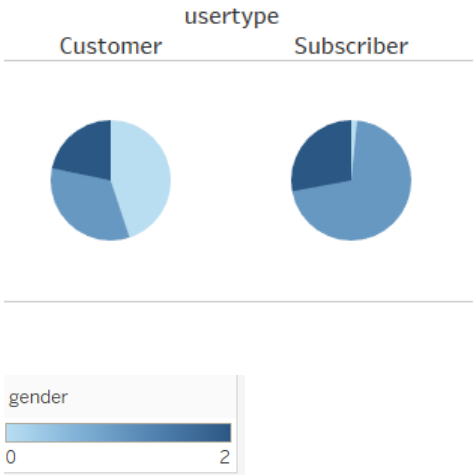
2019 2020

10. User type VS Gender:

- 2019:
  - Here we can see that the gender ratio in no. of customers is about equal i.e. equal male and female. But in Subscriber, there are more males than female. So, we can infer that the people who subscribe to our services annually are males.
- 2020:
  - It continues to be in same ratio even in the following year.



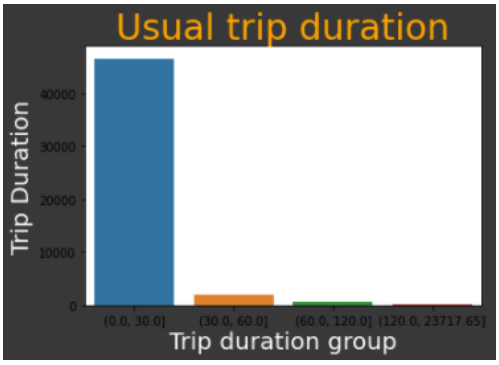
2019



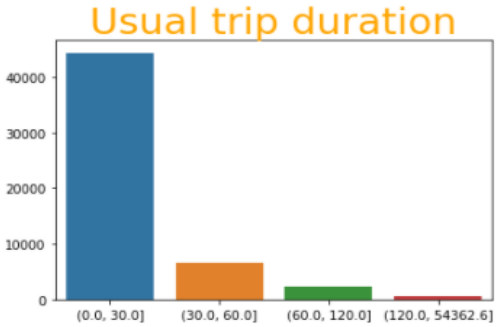
2020

11. Trip Duration:

- We can see that a greater number of people are taking short trips in 2020 than in 2019.

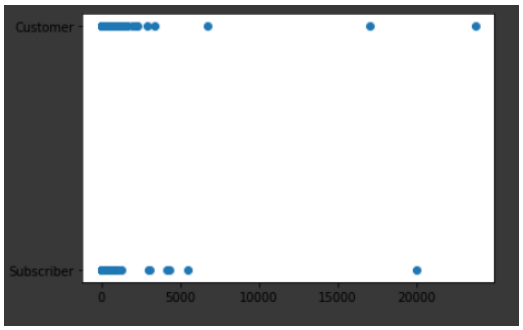


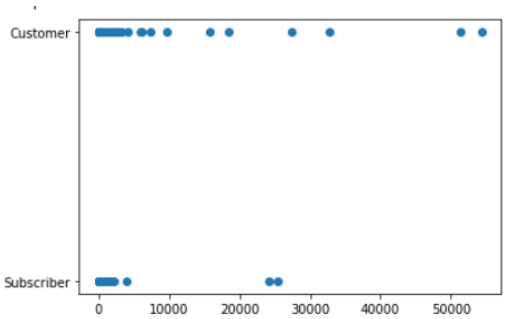
2019



2020

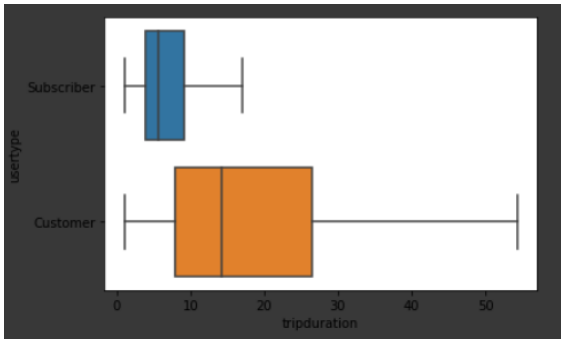
12. Trip Duration as per user type:
    - We can see that in both cases more no. of people uses the service for less time.
    - Customers might use the service for longer periods but subscribers use it for only shorter durations

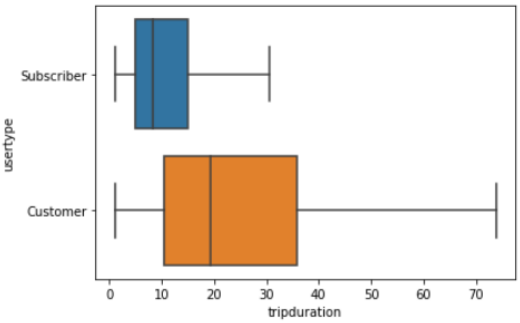




2019 2020

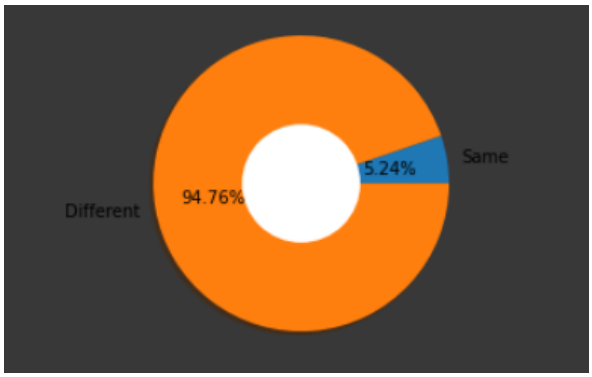
13. Trip Duration of user type without the outliers:
    - We can see that when we exclude the outliers from the data, the median is changing.
    - In 2019:
        - ✓ The median for subscriber is about 5 mins and that for customer is 14.
    - In 2020:
        - ✓ The median for subscriber is about 9 mins and that for customer is 20.
    - We can say that average trip duration in both cases have increased.



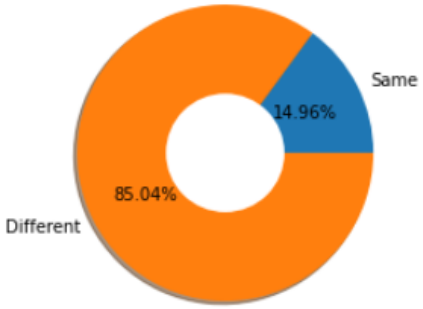


2019 2020

14. Same start and end location VS Different start and end location:
    - We can see that in 2019, people who started from one station mostly ended the trip at a different station.
    - In 2020, more no. of people started the trip and ended the rip at the same station.
    - This might be because of people trying to cycle just to roam around and have fun since they have been staying home for months due to the lockdown.

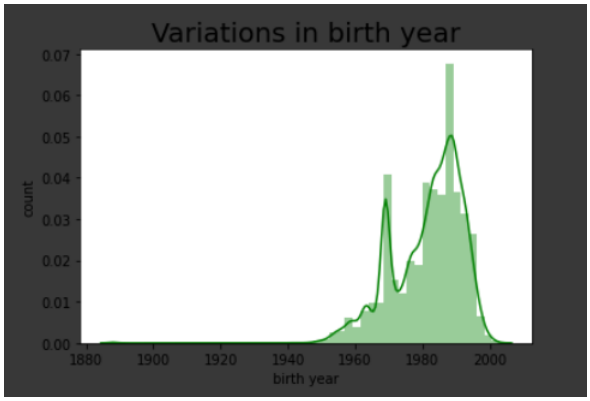


2019

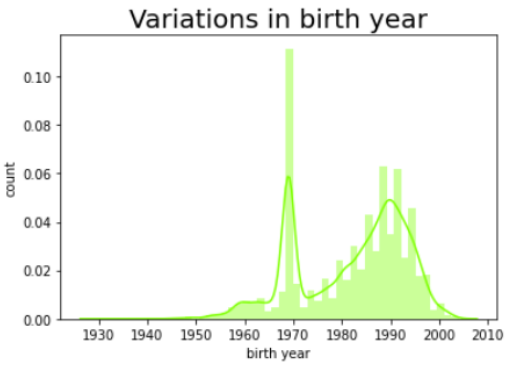


2020

15. Users by their birth year:
    - In 2019, more no. of users was from 1990 but in 2020, we see that the users were more from 1970.
    - This might be because ease of the restrictions and opening of the offices.



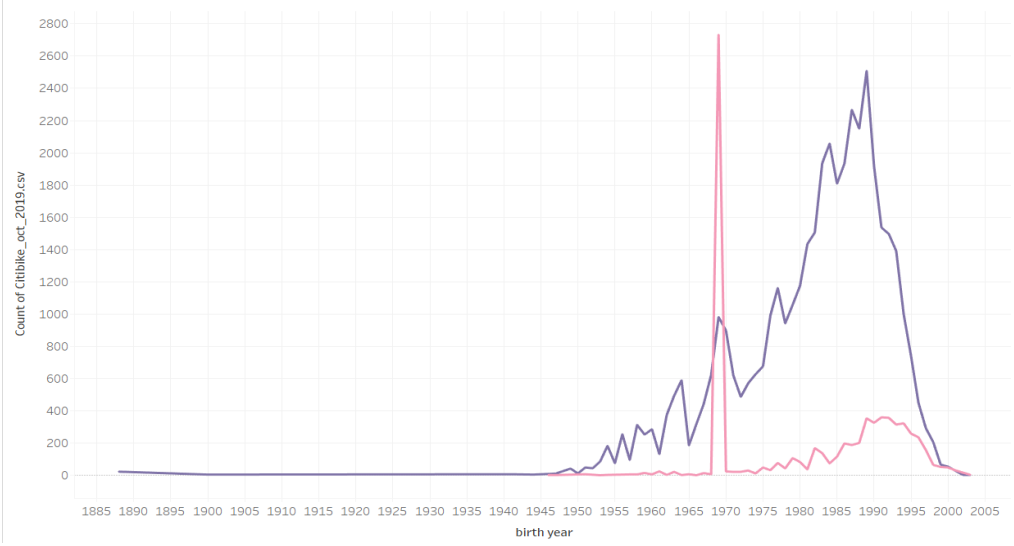
2019



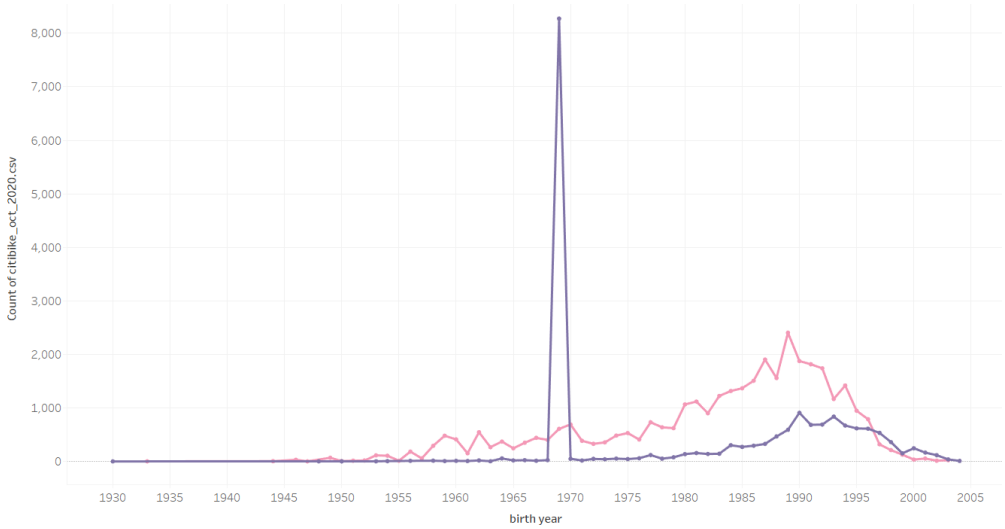
2020

16. User type by their birth year:

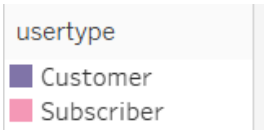
- We can see the trend here and infer that there is a shift in the age group in the following years.
- There are more subscribers from younger generations and there are more customers from the older generations.



2019

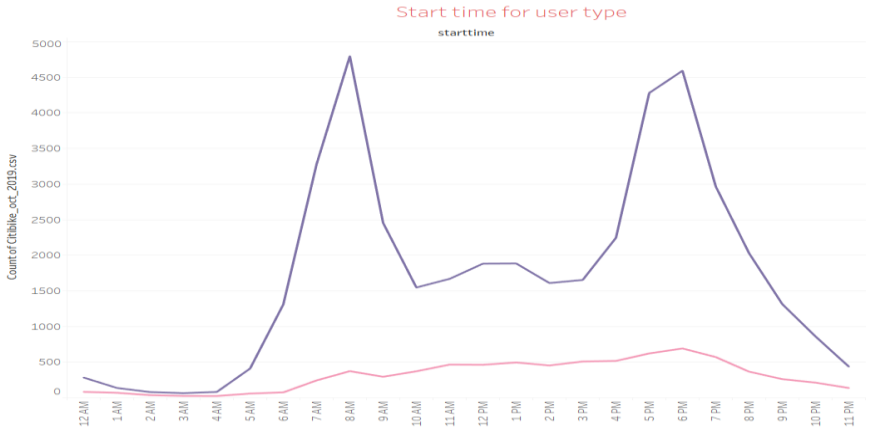


2020

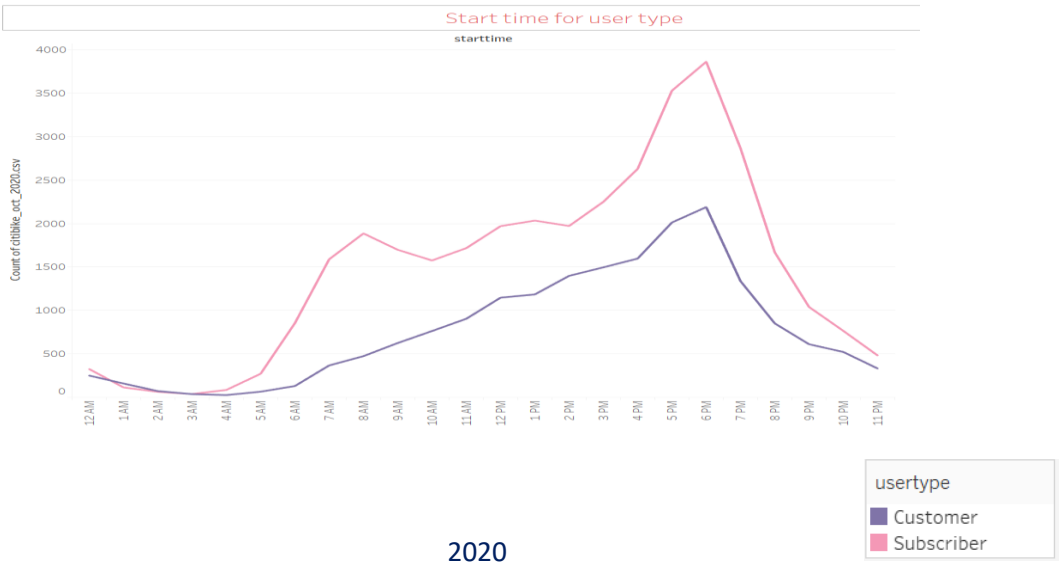


17. Start time by User Type:

- 2019:
  - ✓ More no. of people took trip in the morning (8-9 am) and evening time (6-7 pm).
  - ✓ The subscribers follow a different linearity. Since subscribers are mostly tourists, they tend to use the service more in the afternoon and the evening time.
- 2020:
  - ✓ More no. of people took trip in the evening time (5-7 pm).
  - ✓ Since the offices are partially closed and more no. of corporate people are doing work from home, so there is no morning peak user time.
  - ✓ Users might use the service in the evening as a part of their exercise or just to have fresh air a we can see from the above inference that more users are having same stop stations.
  - ✓ We can also see in the user type, that both are having a similar trend which might means that some customers might have turned to subscription due to uncertainty.



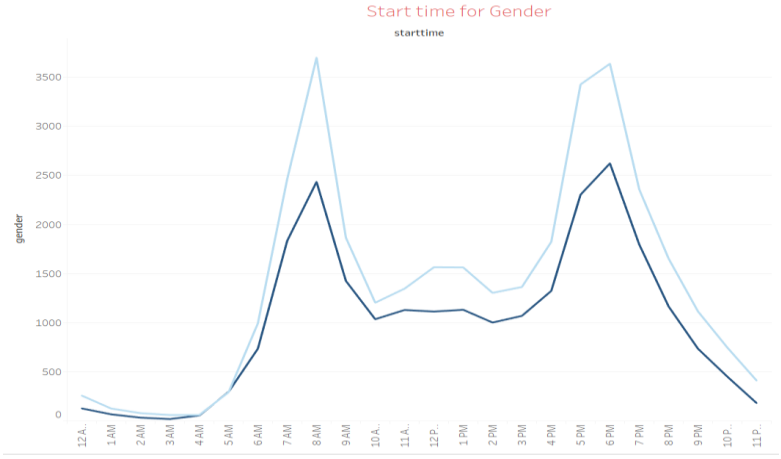
2019



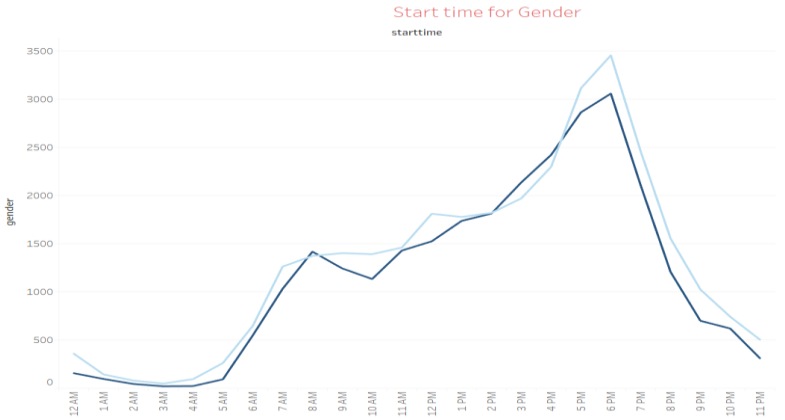
2020

18. Start time by Gender:

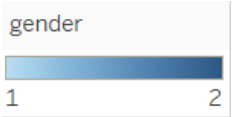
- We can see that in both years, the trend is similar in terms of gender. Both user type are having peak consumption at the same time.



2019

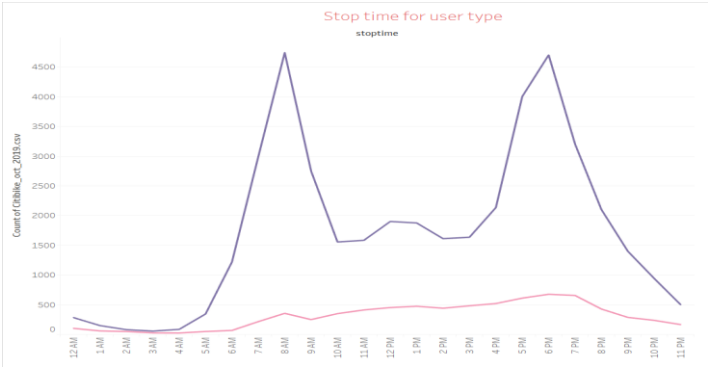


2020

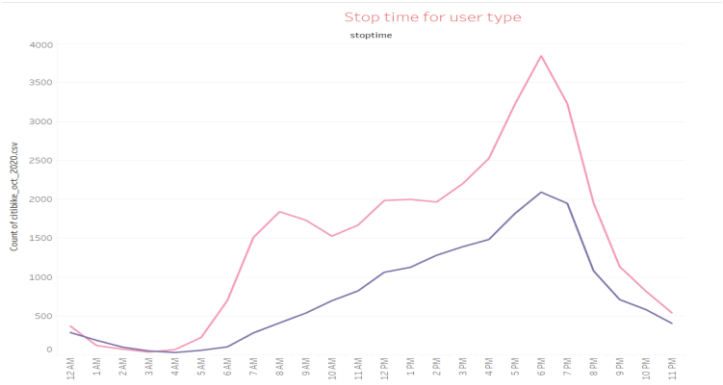


19. Stop time for User Type:

- 2019:
  - ✓ More no. of people end trip in the morning (8-9 am) and evening time (6-7 pm).
  - ✓ The subscribers follow a different linearity. Since subscribers are mostly tourists, they tend to use the service more in the afternoon and the evening time.
- 2020:
  - ✓ More no. of people end trip in the evening time (5-7 pm).
  - ✓ Since the offices are partially closed and more no. of corporate people are doing work from home, so there is no morning peak user time.
  - ✓ Users might use the service in the evening as a part of their exercise or just to have fresh air a we can see from the above inference that more users are having same stop stations.
  - ✓ We can also see in the user type, that both are having a similar trend which might means that some customers might have turned to subscription due to uncertainty.
- We can also see the trend could be because of users taking trips for less than 30 minutes.



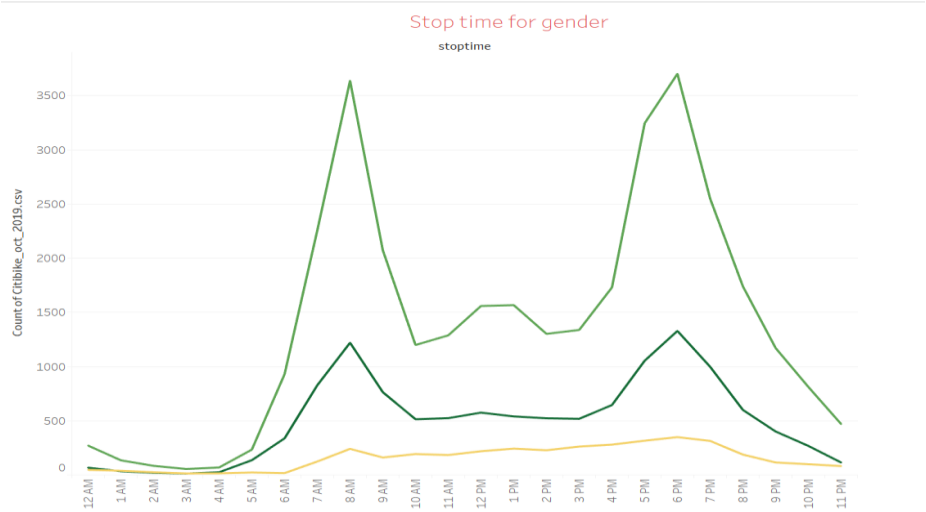
2019



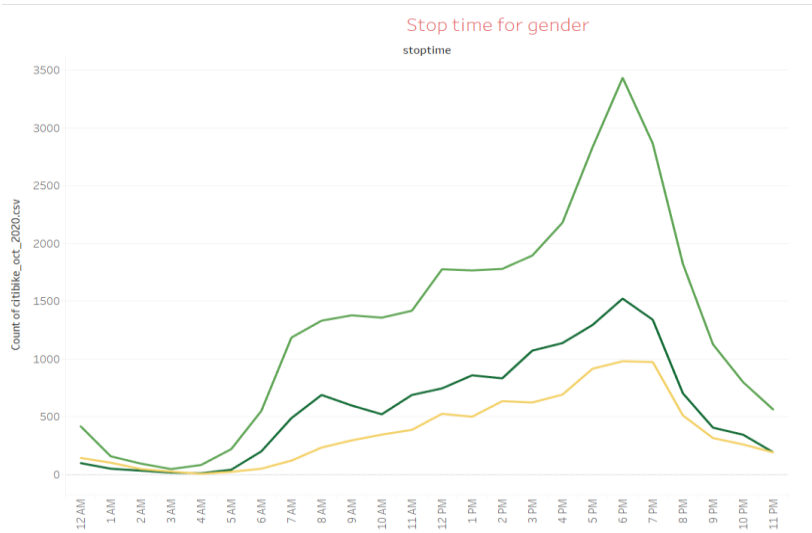
2020

usertype
Customer
Subscriber

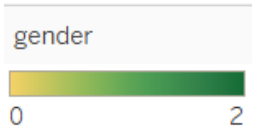
20. Stop time:



2019

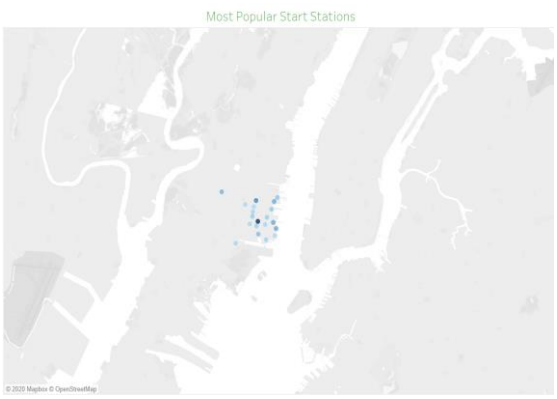


2020

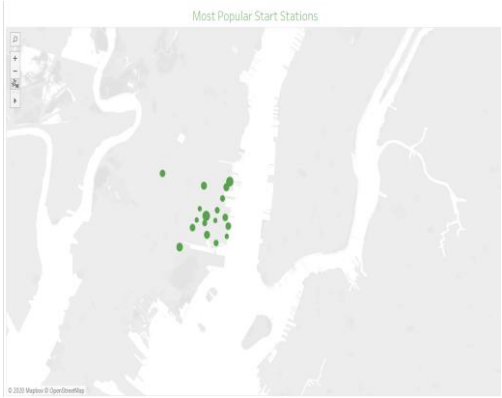


21. Trip count:
    - To show arrival and departure at any station on a particular time.

22. Popular Start Stations on Map:



2019



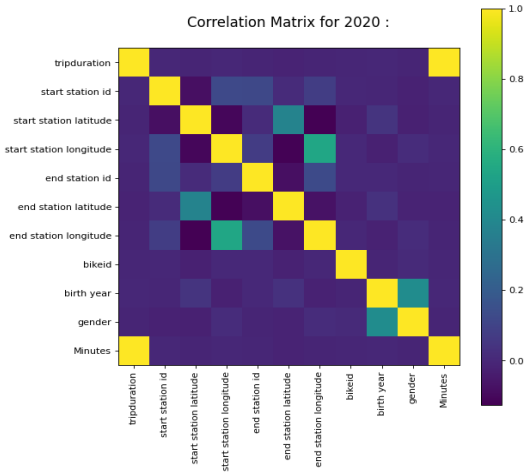
2020

23. Popular End Stations on Map:

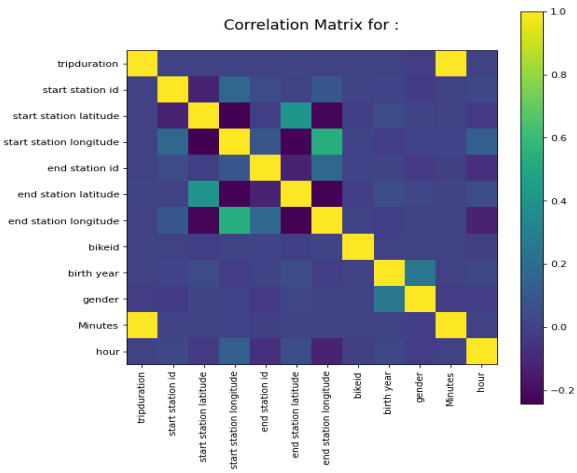
2019

2020

24. Correlation Matrix:

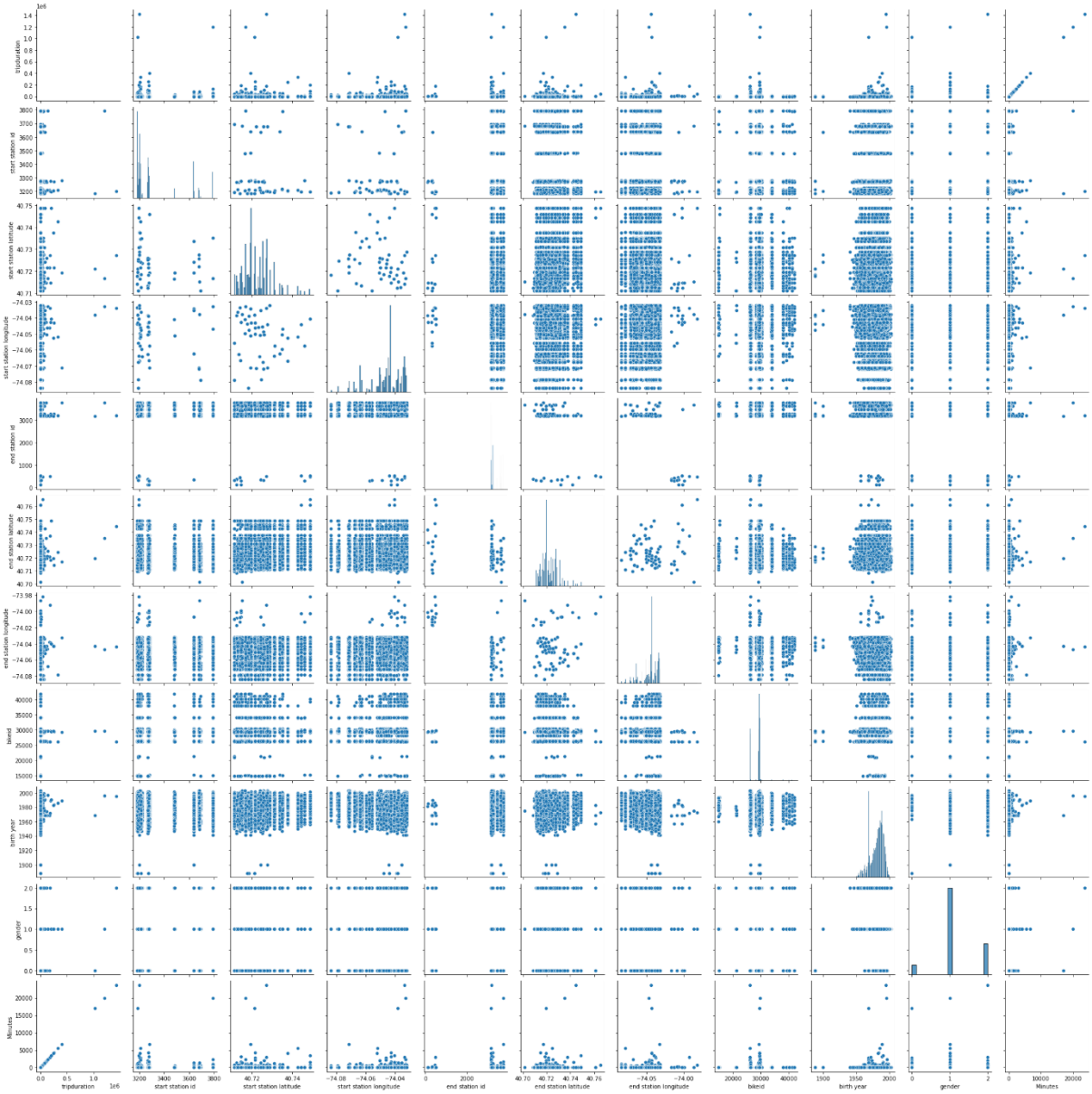


2019



2020

25. Pair Plot:



2019

tripduration
start station id
start station latitude
start station longitude
end station id
end station latitude
end station longitude
bikeid
birth year
gender
Minutes

2020

Conclusions:

1. There is increase is no. of cycles.
2. There is increase in no. of end stations.
3. The busiest station is Grove street path even in 2020.
4. The top 10 start stations and end stations over the years have changed.
5. People may take a round trip with citibike.
6. There are stations where People may never start the trip but end the trip.
7. Males use the service more often than any other gender (in both years).
8. People subscribing to service has decreased and there is increase in no. of customers over the years.
9. Customers are equal in terms of gender ratio but in subscribers, males are dominant.
10. More people are taking longer trips over the years.
11. Overall, the trip time is increased for both user types.
12. More users are starting and ending the trip at the same station.
13. We can see that the user age group has shifted over the year.
14. The peak time has changed over the years.

usertype	tripduration	birth year
Customer	36.902127	1981.280049
Subscriber	9.387956	1981.653455

usertype	tripduration	birth year
Customer	47.506123	1980.744946
Subscriber	15.095522	1982.658542

Conclusion:

Which type of customer travels longer, on average? -Customers (those without subsciption)

Which type of customer sees greater variance in trip duration? -Customers.

Which type of customer sees greater outlier data points in trip duration? -Customers.

Why is that?

In general, annual members are usually New York locals and 3-day pass tickets are targeted for tourists or visitors.

They tend to make the most of their day pass, which leads to longer travel time.

On the other hand, annual subscribers don't bother to make a long trip because they know they will use it more frequently.

Suggestions:

- Overall, we can say that there is increase in growth of the company but the pandemic has affected the growth.
- The expansion of services might work but would be a liability at this point of time. So, the stations where there is lower no. of users, services must be halted for a while.
- Since most of our users are males, we must target female users to subscribe to our service. We can offer them lower prices and make awareness campaigns for healthy life by cycling (exercising).
- Since more users are born in year 1980 i.e. in the range 35-40 years, we can target the younger age groups to use our services.

Future Scope:

As we know about the user peak time, the timings, start stations, stop stations, trip duration, etc, we can prepare an ML algorithm to predict the user influx at any given time and deploy cycles to the given station to supply the demand and have uninterrupted services.